

Boosting CAR T cell functionality with oncolytic viruses for the treatment of pediatric diffuse midline gliomas

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Despite the success of CAR (chimeric antigen receptor) T cells in hematological malignancies, their effectiveness against solid or brain tumors, such as pediatric diffuse midline gliomas (DMGs), is limited. CAR T cell success is hampered by factors including immunosuppression from DMGs and their surrounding tumor microenvironment (TME). Oncolytic viruses (OVs) can reverse this immunosuppression, suggesting a potential combination with CAR T cells. Here, we show that infection with Goravir adenovirus and R124 reovirus induced DMG cell lysis ($n = 6$ cultures), with minimal effect on the viability of B7H3- or GD2-targeted CAR T cells, even at high virus concentrations. In addition, RNA sequencing of infected tumor cells revealed altered gene expression in cell cycle and antiviral response pathways. Furthermore, co-cultures of CAR T cells with OV-infected DMGs enhanced CAR T-specific anti-tumor killing in 14 out of 24 cases. The successful combinations exhibited enhanced cytokine and chemokine release, coupled with an increased cytotoxic phenotype. These findings highlight the benefit of DMG pre-infection with OVs to boost CAR T cell activity and suggest that immune stimulation is a key driver of enhanced combination responses.

INTRODUCTION

Pediatric diffuse midline gliomas (DMGs) are primary brain tumors that usually develop in the pons of the brainstem, although these tumors may also arise in the thalamus, spinal cord, or cerebellum.¹ DMGs are characterized and divided into several subtypes based on their lysine 27-to-methionine mutation in genes encoding histone 3 protein (H3K27M), which acts as the driver of DMG initiation and development.² While DMGs have a very poor prognosis, radiotherapy has been the only proven treatment so far, providing some prolongation of survival up to 11 months after diagnosis, highlighting the need for more effective therapies.^{3,4}

Chimeric antigen receptor (CAR) T cell therapies have significantly improved the survival of hematological malignancies.^{5,6} Further-

more, CAR T cells targeting the B7 homolog 3 (B7H3) protein and disialoganglioside type 2 (GD2) have demonstrated early signs of effectiveness in a subgroup of patients with DMG (B7H3-CAR Ts: NCT04185038, GD2: NCT04196413).^{7,8} However, DMGs have an immune-suppressive tumor microenvironment (TME)⁹ that leads to reduced efficacy of CAR T cell therapies.¹⁰ The effect of this immunosuppressive TME remains a challenge for cellular immunotherapies, and several interventions are under investigation to circumvent and improve CAR T cell responses.¹¹ One such strategy that shows promise is the use of oncolytic viruses (OVs).¹² OVs can target and kill tumor cells and have shown promise in cancer treatment.¹² Currently, many different species of OVs, such as adenoviruses and reoviruses, are being investigated for cancer therapy. Each OV species can target tumor cells with specific characteristics across a range of tumor types, making them potent candidates for broader selection of tumors and for personalized treatment.¹³ In addition, preclinical investigations and clinical trials have demonstrated the role of OVs as immune stimulants, altering the TME, thus potentiating a long-term anti-tumor response.^{12,14,15} As such, studies combining OVs and CAR T cells hint at various factors that could lead to successful therapeutic combinations.¹⁶

In this study, we investigate the potential of combining two OVs, the human adenovirus B Goravir¹⁷ (Goravir) and type 3 mammalian orthoreovirus R124¹⁸ (R124), with CAR T cells targeting the two most promising DMG targets described in the literature, B7H3 and GD2.^{7,8} We analyzed the cytotoxicity and effector state of CAR T cells induced by co-incubation with virus-infected cultures. Six patient-derived DMG cultures were analyzed *in vitro* to investigate the

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combinatorial potential of each OV in combination with CAR T cells targeting B7H3 or GD2.

RESULTS

CAR T cells demonstrate high tolerance to OVs and greatly restrict viral infection and replication

The tumor specificity of both Goravir and R124 was tested across six DMG cultures, B7H3- and GD2-CAR T cells, and their untransduced matched donor T cells (UTD), by assessing the ability of the OVs to infect, replicate, and lyse the various cells. Flow cytometry experiments demonstrated more than 30% infection of DMG cells after 24 h of exposure to 20 multiplicity of infection (MOI) of Goravir, as assessed by anti-Hexon staining (Figure 1A). This percentage increased to over 80% in all six DMG cultures after 72 h of exposure, while exposure of both CAR T and UTD T cells to 20 and 200 MOI of Goravir resulted in fewer than 5% of infected cells even after 72 h (Figure 1A; Figure S1A). Exposure of DMG cultures to 10 MOI R124 also increased the percentage of infected cells over time, as measured by anti- σ 3+ staining (Figure 1B). Here, 10 and 100 MOI R124 also infected CAR T and UTD T cells at 24 h, but the percentage of infected cells decreased after 72 h, indicating a failure of sustained R124 infection (Figure 1B; Figure S1B). These results were confirmed by viral titer measurements, as quantified by qPCR (Figures 1C and 1D). While the viral titer of Goravir and R124, as quantified by qPCR, increased from 24 to 72 h in DMG cultures, the replication of both OVs remained limited in all CAR T and UTD T cell cultures, resulting in significantly lower viral titers (Figures 1C, 1D, S1C, and S1D).

Next, DMG, CAR T, and UTD T cell cultures were challenged with increasing concentrations of Goravir or R124 for 72 h to evaluate the lytic effect of these OVs using the Cell Titer-Glo cytotoxicity assay. Goravir infection induced a dose-dependent oncolytic effect in five of six DMG cultures. JHH-DIPG-01, HSJD-DIPG-07, and VUMC-DIPG-10 cell viability decreased by more than 40%, while VUMC-DIPG-11 showed less than a 10% decrease (Figure 1E). DMG cell viability also decreased in a dose-dependent manner after R124 infection, with HSJD-DIPG-07 and VUMC-DIPG-11 decreasing by more than 30% and the other four DMG cultures remaining above 85% cell viability after infection with 200 MOI of R124 (Figure 1F). UTD T cells experienced a viability decrease of more than 30% after exposure to 400 MOI of Goravir or 200 MOI of R124 (Figures 1E and 1F). In contrast, the viability of B7H3 and GD2 CAR T cells remained unaffected, even at increasing MOIs of Goravir and R124 (Figures 1E and 1F). Thus, despite the OV's ability to specifically replicate only in DMG cells, high concentrations of OVs led to cytotoxicity of non-tumor cells, with CAR T cells tolerating these concentrations better than UTD T cells. We therefore selected MOIs of 200 for Goravir and 100 for R124 for further experiments to prevent potential cytotoxic effects on T cells.

Goravir downregulates DMG cell cycle and growth pathways, while R124 upregulates interferon-stimulated genes

Next, we collected mRNA from DMG cultures infected with each OV for 24 and 72 h, as well as from uninfected DMG controls, for bulk

RNA sequencing to define infection-related cellular responses. Unsupervised clustering of the samples resulted in grouping based on the molecular subtype (histone 3 wild-type, *H3F3A* mutant, or *HIST1H3B* mutant) of the cultures, highlighting their heterogeneous background² (Figure S2A). After correcting for culture-specific identity bias, recalculated principal component analysis (PCA) revealed unsupervised clustering that was independent of the culture of origin (Figure S2B). Instead, samples grouped according to infection status, with Goravir-infected and R124-infected DMGs forming distinct clusters, indicating that each OV induces a unique transcriptomic change over time (Figure S2C).

Supervised differential expression analysis subsequently resulted in 3,407 significantly upregulated ($\log_2$ fold change >1 and adjusted p value < 0.05) and 2,715 downregulated ($\log_2$ fold change < -1 and adjusted p value < 0.05) genes in Goravir-infected DMGs compared to uninfected DMGs (Figure 2A; Table S1A). Differentially expressed genes (DEGs) with altered expression were correlated with stress response (*GADD45B*, *MXD3*, and *HMGB2*), whereas cell cycle-associated genes (*CDK4*, *MKI67*, and *CDKN1B*) were mostly downregulated (Table S1A). Infection of DMGs with R124, however, induced the upregulation of 18 DEGs (Figure 2B; Table S1B). However, unlike Goravir, the transcriptomic changes specifically focused on genes such as *MX1*, *ISG15*, and *IFIT3*, which are involved in anti-viral response (Figure 2B).

Reactome analysis, using the DEGs from each OV infection, demonstrated the different pathways each OV manipulated during DMG infection. The DEGs discovered during Goravir infection were enriched in terms such as “cell cycle checkpoints,” “DNA double-strand break repair,” and “small ubiquitin-related modifier (SUMO)ylation” (Figure 2C; Table S2A). Reactome pathway analysis also showed that the 18 DEGs identified during R124 infection were enriched in terms such as “interferon (IFN) signaling,” “Interferon-stimulated genes (ISG)15 antiviral mechanism,” and “protein kinase R (PKR)-mediated signaling” (Figure 2D; Table S2B). These results indicate that cell cycle arrest and stress response are the main pathways affected during Goravir infection, while R124 infection modulates the induction of type I and II IFN responses. We therefore hypothesized that, although the two OVs act via different mechanisms, both may induce an improved effect in combination with CAR T cells.

CAR T cell-specific anti-tumor cytotoxicity is enhanced by pre-infecting DMGs with OVs

To assess the effect of virus infection on CAR T cell efficacy, we first determined the cytotoxicity of B7H3- and GD2-CAR T cells against the different DMG cultures. Immunofluorescent imaging confirmed the presence of both epitopes on the membrane of DMG cells (six out of six positive for B7H3 and five out of six positive for GD2), with B7H3 showing uniform expression and GD2 showing more variation, especially in the case of HSJD-DIPG-07 (Figure S3A and S3B). Whereas VUMC-DIPG-10 cells were GD2 negative, as previously described¹⁹ (Figures S3A and S3B). When co-cultured for 48 h, CAR T cells induced dose-dependent cytotoxicity with varying

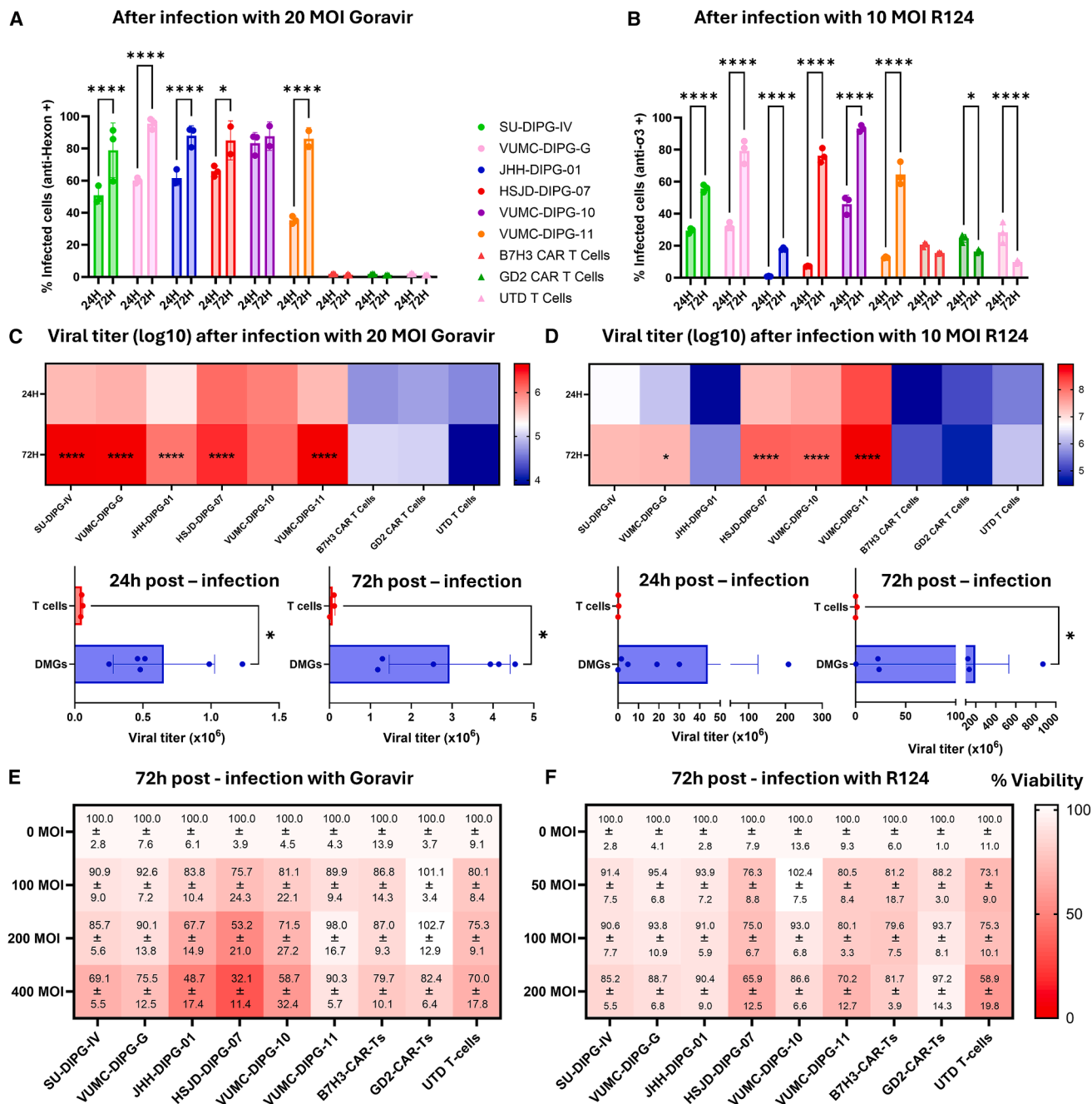


Figure 1. Goravir and R124 induce cytotoxicity through infection and tumor-restricted replication

Patient-derived diffuse midline glioma (DMG) cultures, CAR T cells, and untransduced T cells (UTD) were incubated with the indicated multiplicity of infection (MOI) of Goravir or R124 for 24 and 72 h. Comparisons of A) Goravir (anti-hexon+) and B) R124 (anti-σ3+) infected cells out of the total viable cell population of each culture after 24 and 72 h of infection are shown. Viral titer comparison of C) Goravir (type B polymerase copies) and D) R124 (S4Q copies) per cell culture between 24 and 72 h of infection and across DMG and T cells is also shown. Viability changes per cell culture after incubation with increasing MOIs of (E) Goravir or (F) R124 are shown. For (A) and (B), each bar and error bar represent the mean ± SD of % infected cells ($n = 3$ biological replicates). Statistical analysis was measured using unpaired t tests per culture. For (C) and (D), each heatmap box indicates the mean ± SD of log10-transformed viral titer values ($n = 3$ independent experiments), while each bar and error bar represent the mean ± SD of viral titer ($\times 10^6$) ($n = 6$ DMG and $n = 3$ T cell cultures). For (E) and (F), each box indicates the mean ± SD of viability values from CellTiter-Glo experiments ($n = 2$ or 3 independent experiments), normalized to the baseline viability of each untreated sample. The color scale indicates viability per MOI from white (100%) to red (0%). Statistical significance was assessed using unpaired t tests for heatmaps and the Mann-Whitney test for boxplots. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.

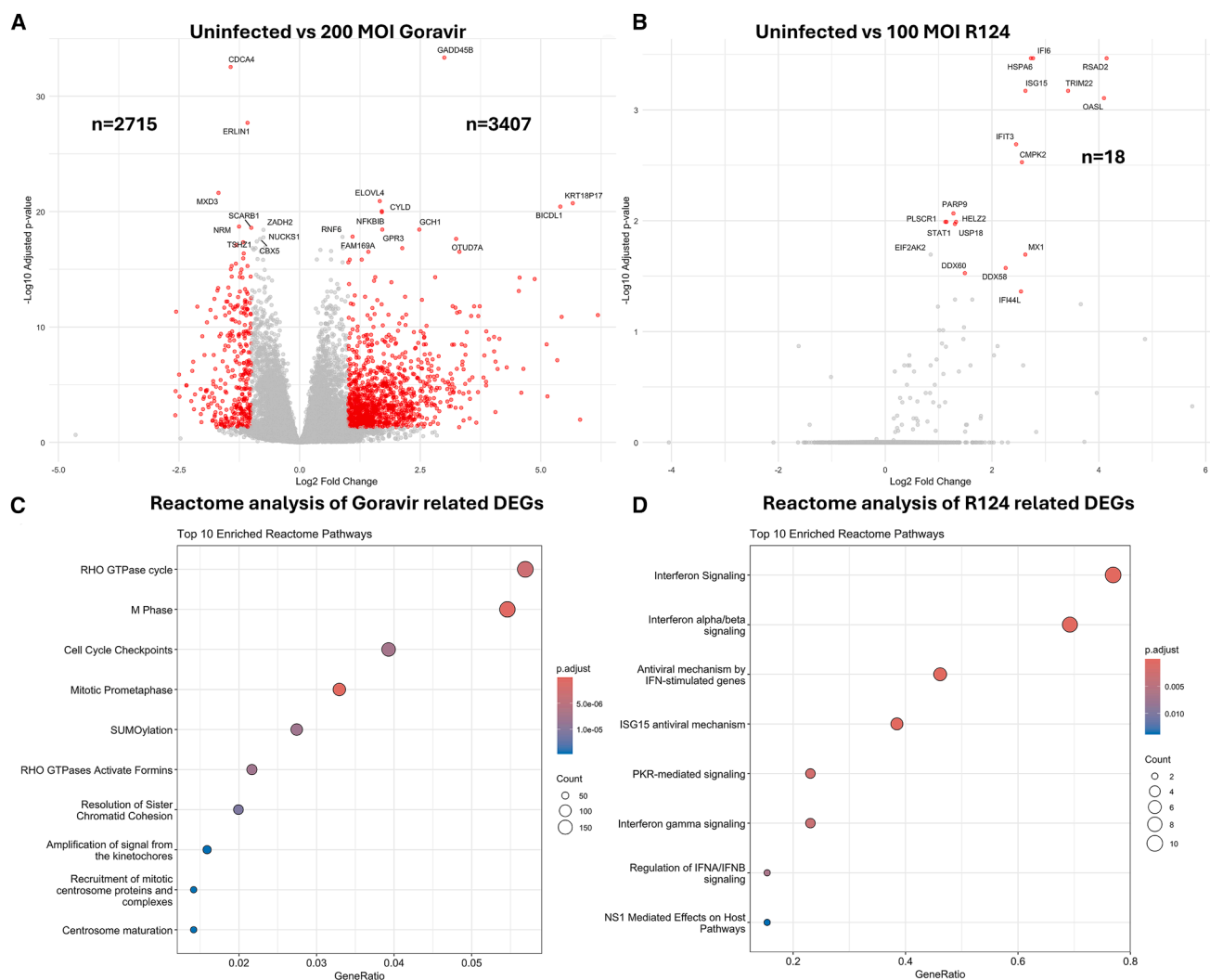
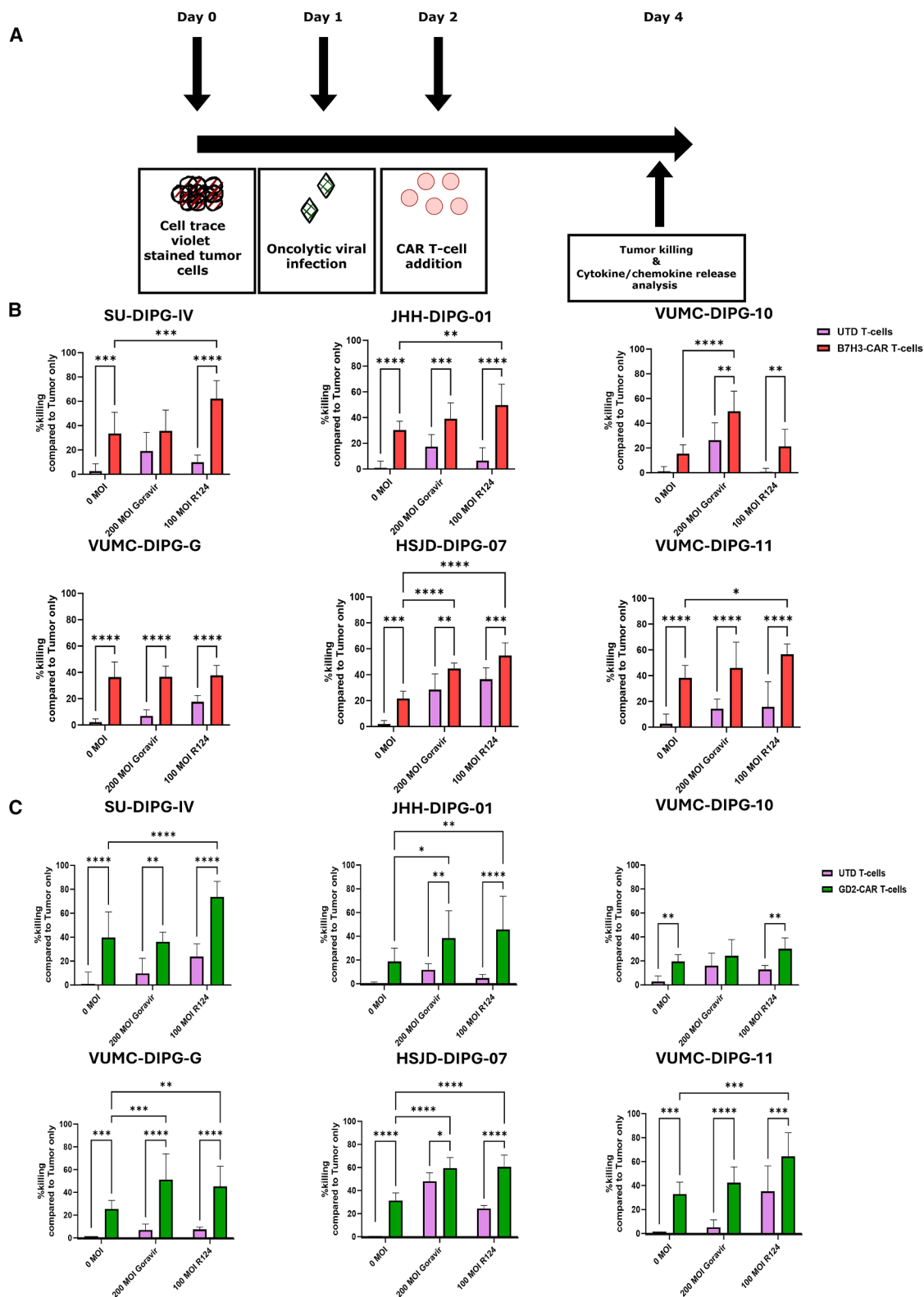


Figure 2. Goravir and R124 infection affect different pathways in DMG cultures

Volcano plot of differentially expressed genes (DEGs) in DMG cultures (SU-DIPG-IV, VUMC-DIPG-G, JHH-DIPG-01, HSJD-DIPG-07, VUMC-DIPG-10, and VUMC-DIPG-11) infected with (A) 200 multiplicity of infection (MOI) Goravir or (B) 100 MOI R124 for 24 ($n = 6$) and 72 h ($n = 6$) compared with uninfected DMG cultures ($n = 18$). Dot plot of the top 10 Reactome pathway analysis of (C) Goravir- or (D) R124 induced DEGs, ordered by adjusted p value, are also shown. For (A) and (B), each dot in the volcano plots represents a gene, with gray genes indicating absolute log2 fold change < 1 or adjusted p value < 0.05, and red genes indicating absolute log2 fold change > 1 and adjusted p value < 0.05; the top 20 genes are labeled.

efficacies against the six DMGs at increasing effector-to-target ratios (E:T) (Figure S3C). As expected, CAR T cells exhibited higher tumor killing and IFN- γ release compared to UTD T cells, which were used as negative controls for epitope-specific killing (Figure S3D and S3E). CAR T cells exhibited the lowest killing (<21%) and IFN- γ release (<400 pg/mL for B7H3 CAR and <100 pg/mL for GD2 CAR) upon exposure to and incubation with the B7H3-low and GD2-negative VUMC-DIPG-10 cultures (Figures S3D and S3E), highlighting the CAR-epitope dependency for effective CAR T cell anti-tumor activity. Next, the DMG cultures were pre-infected for 24 h with 200 MOI Goravir or 100 MOI R124 and then co-cultured with either B7H3- or GD2-CAR T cells for 48 h to investigate the combinatorial anti-tu-

mor effect (Figure 3A). For each DMG culture, the maximal E:T ratio that resulted in $\leq 50\%$ tumor cell killing when used as monotherapy was selected to allow assessment of combination effects on DMG killing (Table S3). Indeed, pre-infection of DMGs enhanced CAR T-induced killing of a subset of DMG cultures, as observed by flow cytometry (Figures 3B and 3C). More specifically, improved killing of two of the six Goravir-infected DMG cultures (HSJD-DIPG-07 and VUMC-DIPG-10) was observed when co-cultured with B7H3-CAR T cells (Figure 3B), whereas B7H3-CAR T cells co-cultured with R124-infected DMGs showed an improved response in four out of the six cultures (SU-DIPG-IV, JHH-DIPG-01, HSJD-DIPG-07, and VUMC-DIPG-11) (Figure 3B).



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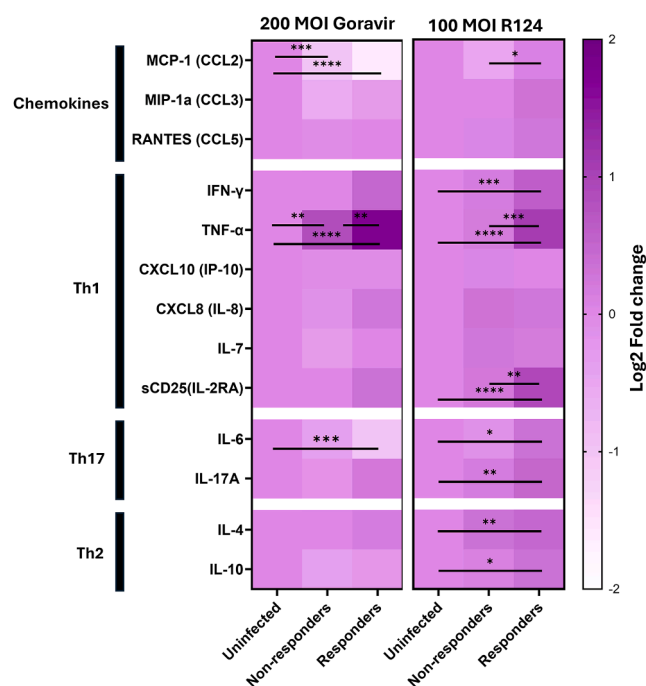


Figure 4. Successful combination responses release increased levels of pro-inflammatory cytokines

Chemokine and cytokine changes in co-culture supernatants from Figures 3B and 3C were grouped into uninfected, Responder, and Non-Responder categories for each OV combination based on Table 1. Data are presented as the mean log₂ (fold change) measured from B7H3 ($n = 6$) and GD2 ($n = 6$) co-cultures of DMG (SU-DIPG-IV, VUMC-DIPG-G, JHH-DIPG-01, HSJD-DIPG-07, VUMC-DIPG-10, and VUMC-DIPG-11) per condition (uninfected, Goravir pre-infected, and R124 pre-infected), using the supernatant from uninfected DMG co-cultures with the respective DMG and CAR T cells as the baseline. The color scale indicates log₂ (fold change) ranging from the lowest (white) to the highest (purple). Statistical analysis was measured using repeated-measures one-way ANOVA for each chemokine/cytokine, with Tukey's test for multiple comparisons. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.

When OV-infected DMGs were co-cultured with GD2-CAR T cells, more successful combinations were observed (Figure 3C), with GD2-CAR T cell co-cultures resulting in increased killing of three out of the six Goravir-infected DMG cultures (VUMC-DIPG-G, JHH-DIPG-01, and HSJD-DIPG-07) and five out of the six DMG cultures when pre-infected with R124 (SU-DIPG-IV, VUMC-DIPG-G, JHH-DIPG-01, HSJD-DIPG-07, and VUMC-DIPG-11) (Figure 3C). In addition, pre-infection with either Goravir or R124 did not lead to increased killing of the GD2-negative VUMC-DIPG-10 (Figure 3C), indicating the specificity of this combination. This claim was further strengthened by the absence of increased tumor killing

when UTD T cells were co-cultured with either Goravir- or R124-infected DMGs (Figures S4A and S4B).

Further investigation of B7H3 and GD2 expression by flow cytometry after 24 and 72 h of Goravir or R124 infection demonstrated changes in B7H3 and GD2 only after 72 h across various DMGs (Figures S5A and S5B). More specifically, 72 h of Goravir infection led to a reduction of B7H3 (−0.6-fold change) and GD2 (−0.2-fold change) on SU-DIPG-IV and a decrease of B7H3 (−0.3-fold change) on VUMC-DIPG-10 (Figures S5A and S5B). However, a 72-h Goravir infection led to a significant increase in GD2 on VUMC-DIPG-G, JHH-DIPG-01, HSJD-DIPG-07, and VUMC-DIPG-11. R124 infection did not affect either B7H3 or GD2 expression at either time point, except for a GD2 decrease (−0.3-fold change) on SU-DIPG-IV (Figures S5A and S5B).

Taken together, these results demonstrate that the OVs Goravir and R124 enhanced the anti-tumor cytotoxicity of B7H3- and GD2-CAR T cells against most of the DMG cultures tested, independently of changes in epitope expression. Notably, R124 enhanced CAR T cell cytotoxicity in more cases than Goravir, while GD2-CAR T cell combinations benefited the most from both OVs.

Increased release of cytokines is related to a successful anti-tumor response

To determine whether the enhanced anti-tumor response observed after OV pre-infection correlated with enhanced cytokine release, we quantified multiple cytokines and chemokines, such as MCP-1 (CCL2), MIP-1a (CCL3), RANTES (CCL5), IFN- γ , TNF- α , CXCL10 (IP-10), CXCL8 (IL-8), IL-7, soluble CD25 (IL-2RA), IL-6, IL-17 A, IL-4, and IL-10, in the supernatant of the pre-infected and uninfected co-cultures. Each B7H3 and GD2 co-culture produced a very heterogeneous cytokine profile across the different uninfected and pre-infected DMG cultures, although general patterns could be observed (Figure 4; Figure S6; Table S4A). Cytokine and chemokine production was also assessed in supernatants from uninfected and OV-infected DMG monocultures (Table S4B). While glioma cells are known to produce cytokines and chemokines,^{20,21} the concentrations detected in the DMG monocultures were considerably lower than those measured in the CAR T cell co-cultures (Tables S4A and S4B), indicating that the majority of cytokines and chemokines detected in the co-culture conditions were primarily produced by the CAR T cells.

Grouping of pre-infected co-cultures that successfully enhanced tumor killing (Responders) and pre-infected co-cultures that failed to enhance tumor killing (Non-Responders) (Table 1) revealed significant differences across these two groups and the uninfected controls (Figure 4). More specifically, significant differences (adjusted

Figure 3. OV pre-infection of DMGs enhances B7H3 and GD2 CAR T cell anti-tumor killing *in vitro*

(A) Schematic of the experimental set-up. (B) Anti-tumor activity of either B7H3 CAR T cells or untransduced T cells (UTD), alone or in combination with Goravir or R124, as assessed in *in vitro* co-culture assays. (C) Anti-tumor activity of GD2 CAR T cells or untransduced T cells (UTD), alone or in combination with Goravir or R124, as assessed in *in vitro* co-culture assays. For (B) and (C), each bar and error bar represents the mean $\pm$ SD of $n = 6$ values. Statistical analysis was assessed using two-way ANOVA with Tukey's test for multiple comparisons. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.

Table 1. Responder and Non-Responder status of each co-culture

CAR-T	DMG culture	B7H3	GD2
Goravir pre-infection	SU-DIPG-IV	Non-Responder	Non-Responder
	VUMC-DIPG-G	Non-Responder	Responder
	JHH-DIPG-01	Non-Responder	Responder
	HSJD-DIPG-07	Responder	Responder
	VUMC-DIPG-10	Responder	Non-Responder
	VUCM-DIPG-11	Non-Responder	Non-Responder
	SU-DIPG-IV	Responder	Responder
R124 pre-infection	VUMC-DIPG-G	Non-Responder	Responder
	JHH-DIPG-01	Responder	Responder
	HSJD-DIPG-07	Responder	Responder
	VUMC-DIPG-10	Non-Responder	Non-Responder
	VUCM-DIPG-11	Responder	Responder

Each DMG co-culture was categorized as either Non-Responders or Responders based on the ability of OV pre-infection of DMGs to significantly enhance the anti-tumor cytotoxicity of B7H3 (Figure 3B) or GD2 CAR T cells (Figure 3C) compared with both monotherapies and combinations with OVs and UTD cells.

p value < 0.05) between Responders and Non-Responders co-cultures were observed only in Th1 cytokines, such as TNF- α and sCD25. TNF- α significantly increased in both Goravir (1.7 log₂ fold change) and R124 pre-infected Responders (1.1 log₂ fold change), and sCD25 increased only in R124 Responders (0.9 log₂ fold change) (Figure 4), with the largest TNF- α and sCD25 log₂ fold changes being especially apparent in Responders of GD2 co-cultures (Figure S6). The Th1 cytokine IFN- γ also increased in R124-Responders, although this difference was only significant when compared to uninfected co-cultures (Figure 4). In addition, R124-Responders showed enhanced release of both Th17 cytokines IL-6 (0.4 log₂ fold change) and IL-17 A (0.5 log₂ fold change), as well as Th2 cytokines IL-4 (0.4 log₂ fold change) and IL-10 (0.3 log₂ fold change), only when compared to uninfected controls (Figure 4). Goravir Responders demonstrated a significant reduction of CCL2 (−1.6 log₂ fold change) and IL-6 (−1.0 log₂ fold change) compared to uninfected controls (Figure 4). Of note, in these cases, Goravir and R124 Non-Responders demonstrated a similar pattern of cytokine changes to the Responders, although on a smaller scale.

These results demonstrate that the enhanced anti-tumor response induced by OV pre-infection is correlated with increased stimulation and cytokine release by B7H3- and GD2 CAR T-cells. Additionally, the higher number of cytokine changes observed in Responders pre-infected with R124 further supports the superior immunostimulatory effects of R124 compared to Goravir.

Successful combination therapy is supported by increased cytotoxic and memory-related phenotypic changes of CD4 and CD8 CAR T cells

Finally, we determined whether pre-infection of DMGs with either Goravir or R124 affected markers related to activation, exhaustion,

migration, cytotoxicity, memory, and regulation in both CD4 and CD8 CAR T cells. Spectral cytometry analysis of 22 markers showed OV-dependent changes in the phenotypes of both CD4 and CD8 CAR T cells. At least 10 markers were differentially expressed (adjusted p value < 0.05) in each OV group compared to uninfected co-cultures (Figure 5A). These alterations were more pronounced in OV-Responder CAR T cells than in Non-Responder cells (Figure 5A).

In detail, the activation markers CD69 and HLA-DR were both up-regulated in CD4 and CD8 CAR T cells from Goravir and R124 Responders compared to CAR T cells from uninfected co-cultures (Figure 5A; Figure S7A). The increase in CD69 was statistically significant only in Goravir Responders and Non-Responders (Figure 5A; Figure S7A). CD25 levels also significantly increased on both CD4 and CD8 cells in Goravir Responders but decreased in the CD8 population of R124 Responders (Figure 5A; Figure S7A). Conversely, CD137 levels decreased across all Responders, with the largest reduction observed in both CD4 and CD8 R124 Responders (Figure 5A; Figure S7A). CAR T cell proliferation, measured by ki-67 expression, showed no difference between OV-infected and uninfected co-cultures (Figure 5A; Figure S7B). However, exhaustion markers such as TIM-3, TIGIT, and CTLA-4 increased in Goravir and R124 Responders, while PD-1 and LAG-3 levels remained largely unchanged compared to uninfected controls (Figure 5A; Figure S7C).

Furthermore, CD4 and CD8 CAR T cells from OV Responders mainly demonstrated increased expression of the migration-associated markers CCR9, CCR4, and CD29, whereas CD49d was decreased only in CD4 R124 Responders (Figure 5A; Figure S7D). OV Responders also demonstrated elevated levels of the cytotoxic markers CD56 and CXCR3, with CD95 increased only in Goravir Responders and GrzmB remaining unchanged across OV Responders (Figure 5A; Figure S7E). Additionally, overexpression of the memory-associated markers CCR7 and CD127 was observed in both CD4 and CD8 cells of OV Responders, while the naivety marker CD45RA was reduced in CD4 Goravir Responders (Figure 5A; Figure S7F). No change in the regulatory state of the CAR T cells was observed across the different conditions (Figure 5A; Figure S7G). Despite Responders and Non-Responders demonstrating similar phenotypic changes, these changes were mostly significant in OV Responders compared to the uninfected group (Figures S7A–S7G). However, four markers (TIGIT and CCR7 for Goravir and TIM-3 and CD56 for R124) were significantly elevated in Responders compared to Non-Responders (Figure 5A; Figures S7C, S7E, and S7F).

In line with the increased expression of cytotoxic markers, the % of IFN- γ + cells also increased in CD4 and CD8 CAR T cells of Goravir Responders and in the CD4 population of R124 Responders (Figure 5B). Meanwhile, analysis of the memory state of the CAR T cells demonstrated an increase in the % of effector memory T cells (TEM) populations, in parallel with a decrease in terminally

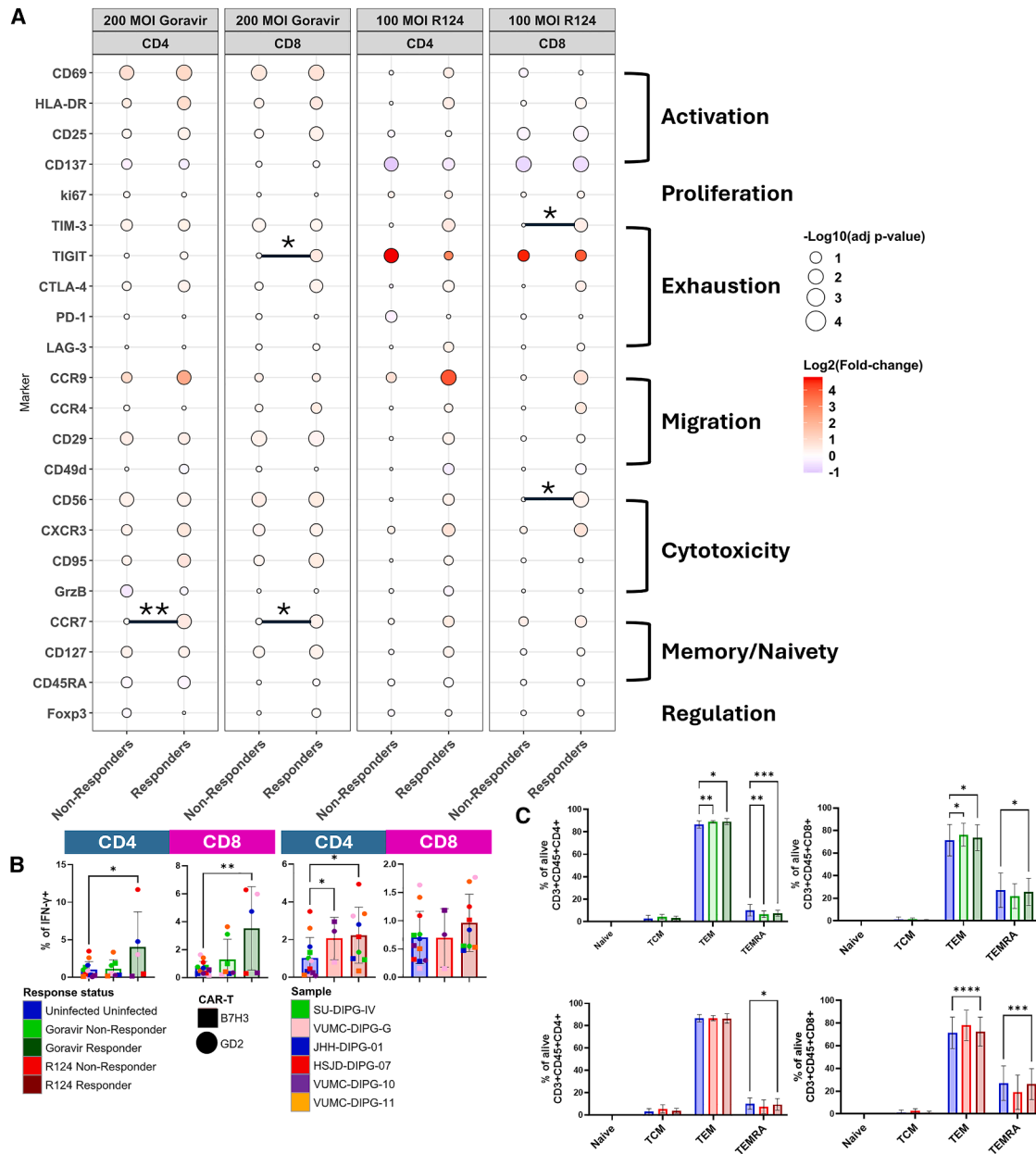


Figure 5. Phenotypic changes in CD4 and CD8 CAR T cells drive successful combination responses

(A) Dot plot demonstrating the log₂ (fold change) of markers associated with T cell activation, proliferation, exhaustion, migration, cytotoxicity, memory/naivety, and regulation, expressed across CD4 and CD8 CAR T cell populations in Goravir Non-Responders and Responder conditions and in R124 Non-Responder and Responder conditions. (B) Dotted bar plots showing population percentages of IFN- γ + CAR T cells from the total population of viable CD3+CD45+CD4+ or viable CD3+CD45+CD8+ cells co-cultured with Goravir- or R124 pre-infected DMGs. (C) Bar plots showing population percentage comparisons of naive T cells (CCR7+CD45RA+), central memory T cells (TCM) (CCR7+CD45RA-), effector memory T cells (TEM) (CCR7-CD45RA-), and terminally differentiated effector memory T cells (TEMRA) (CCR7-CD45RA+) in CD4 (left) and CD8 (right) co-cultures with Goravir (top) or R124 (bottom) pre-infected DMGs. For the dot plot, data are presented as mean log₂ (fold change) measured from B7H3 ($n = 6$) and GD2 ($n = 6$)-co-cultures of DMG (SU-DIPG-IV, VUMC-DIPG-G, JHH-DIPG-01, HSJD-DIPG-07, VUMC-DIPG-10, and VUMC-DIPG-11) per condition (uninfected, Goravir pre-infected, R124 pre-infected), indicated as decreasing log₂ (fold change) (blue), increasing (red), and unaffected (white), using expression in CAR T cells co-cultured with each uninfected DMG as baseline. The size of each dot plot indicates $-\log_{10}$ (adjusted p value), with increasing sizes indicating significant differences compared with the uninfected condition. Additional lines and asterisks were used to indicate significant differences between Responders and Non-Responders. Each box and bar in the bar plots represents mean $\pm$ SD of specific population % from CAR T cell co-cultures, with each point color-coded to represent the population % of B7H3

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differentiated TEM (TEM-RA+) in CD4 and CD8 CAR T cells of OV Responders (Figure 5C).

Collectively, these data indicate that OV-infected DMGs promoted phenotypic remodeling across CD4 and CD8 CAR T cell subsets, with increased expression of TIM-3 and TIGIT (exhaustion), CD56 (cytotoxicity), and CCR7 (memory) being linked to the heightened anti-tumor activity seen in the OV Responder groups.

DISCUSSION

This study demonstrates that a combination of the OVs Goravir or R124 with B7H3- or GD2 CAR T cells can augment the killing of several DMG cultures. Furthermore, infection of DMGs with OVs enhanced the anti-tumor killing induced by CAR T cells, while also affecting multiple facets of CAR T cell function. These changes spanned from increased cytokine release to a population shift toward CAR T cells exhibiting a more cytotoxic phenotype.

OVs can, either naturally or through genetic modification, infect and replicate within tumor cells, causing oncolysis. Both Goravir and R124 infected and replicated efficiently in DMG cells. Importantly, Goravir infection exhibited specificity for DMG cells, whereas R124 also infected CAR T and UTD T cells; however, this broader tropism did not translate into productive replication outside of DMG cells, as viral titers remained stable and the initially infected T cell populations declined over time. Both OVs induced DMG lysis to varying degrees, and despite the high levels of infection and high viral titers after 3 days of infection, R124 demonstrated a relatively modest oncolytic effect against most DMG cultures. This discrepancy probably reflects a time interval between viral replication and the onset of virus-mediated cytotoxicity, as productive infection and viral replication precede cell death.²² Consistent with this biological explanation, we previously observed greater R124-induced DMG lysis using the same DMG cultures and comparably high R124 concentrations at a later time point (day 5 post-infection).¹³ These findings indicate that effective replication and cytolytic activity of both OVs are restricted to DMG cells, although high OV concentrations resulted in reduced viability of UTD T cells and, to a lesser extent, CAR T cells. Thus, these results support the therapeutic potential of Goravir and R124 against DMG cells in the presence of CAR T cells.

RNA analysis showed that DMGs infected with OVs underwent major transcriptional changes centered on key pathways during the process of infection. Notably, Goravir infection downregulated pathways related to the cell cycle and stress responses through SUMO-mediated modifications. These mechanisms have previously been documented for type 5 and 2 adenoviruses,^{23–25} further complementing our findings. R124 infection, however, according to reac-

tome enrichment analysis, resulted in the induction of type I and II IFN responses. These responses are usually upregulated after detection of double-stranded (ds) RNA to limit infection and signal nearby cells through interactions with PKR,²⁶ leading to ISG induction^{22,27} and potentially explaining the lower oncolytic effect observed in these cultures after R124 infection.

Additionally, utilizing six DMG cultures with a diverse molecular background and varying viral infection and replication dynamics, as well as different levels of B7H3 and GD2, acted as surrogates of interpatient variability.^{19,28} This setup allowed us to assess the effectiveness of the viral pre-treatment.¹³ It should be noted that the OV concentrations and E:T ratios used in the combination experiments were intentionally selected to prevent T cell toxicity at high OV concentrations and to maintain submaximal anti-tumor conditions in monotherapy, in order to preserve the dynamic range needed to detect combination effects. As a consequence, our findings revealed that combining OVs with CAR T cells produces different degrees of synergy across DMG cultures, with the level of synergy not correlating with the limited OV-driven changes observed in B7H3 and GD2 expression. The combination of Goravir and B7H3 cells enhanced killing in only two out of six cultures, while the combination of R124 and GD2-CAR T cells enhanced killing in all GD2-expressing DMGs. These results demonstrate that OV pre-infection of either susceptible or resistant DMG cultures can either enhance or fail to enhance the B7H3- and GD2-targeted anti-tumor response of CAR T cells.

Besides the variability in the efficacy of tumor clearance, substantial variability in cytokine production was observed per CAR T cell-DMG co-culture across the different conditions. Yet, stratification of the co-cultures into Responders and Non-Responders based on successful combination treatment showed that TNF- α concentrations were significantly higher in Responders to either virus/CAR T cell combination. A similar effect was seen for sCD25 in Responders to the R124 combination therapy with CAR T cells. TNF- α , in particular, has previously been shown to be produced during viral infections, and ectopic TNF- α overexpression by a genetically modified adenovirus led to enhanced tumor regression and infiltration of both CAR T cells and the host's T cells in pancreatic ductal adenocarcinoma models.²⁹ Despite the ability of Goravir to induce immunogenic cell death in pancreatic cancer cells *in vitro*,³⁰ Goravir induced limited cytokine release changes in Responders compared to R124. As shown by our transcriptomic analysis, the superior immune-stimulating effects of R124 were further supported by the elevated release of additional cytokines, such as IFN- γ , IL-6, IL-17 A, IL-4, and IL-10 in R124 Responders, leading to enhanced anti-tumor killing. These specific cytokines have also been linked to, and correlated with, enhanced CAR

(square) ($n = 18$) or GD2 (circle) ($n = 18$) CAR T cells across uninfected, Non-Responders, and Responder conditions of either Goravir or R124 pre-infected DMG co-cultures. Statistical analysis was performed using repeated-measures one-way ANOVA corrected with Tukey's test for parametric data, while for non-parametric data, the Friedman paired test corrected with Dunn's test for multiple corrections was used. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$. The dot plot was plotted in R (<https://www.r-project.org/>).

T cell-specific anti-tumor responses after OV infection of the tumor.^{31,32} Interestingly, IL-4 has previously been shown to maintain CAR T cell longevity, leading to long-term tumor remission.³³

Spectral cytometry further demonstrated the ability of both Goravir- and R124-infected DMGs to remodel CD4 and CD8 CAR T cell phenotypes, characterized by several phenotypic changes associated with an enhanced combinational response. Particularly, the uncoordinated expression changes of activation-associated markers observed in OV Responder CAR T cells suggest enhanced bystander activation. This pattern is consistent with previous reports demonstrating increased bystander activation in response to released viral peptides and high Th1 cytokine exposure,³⁴ both of which are present in the OV-infected co-cultures. In tandem with the enhanced CAR T cell activation, OV Responders selectively upregulated exhaustion-associated markers, the role of which remains unclear in this setup due to the short duration of the co-culture. These markers are mostly expressed in response to T cell activation and only lead to functional exhaustion after prolonged T cell activation.^{35–38} In addition, the upregulation of migratory, cytotoxicity, and memory markers observed in OV Responders further highlighted the benefit of OV pre-infection on the anti-tumor activity of CAR T cells. These observations are in line with previous reports showing an increase in migration and cytotoxic markers during viral infections^{39–43} and have also been linked with improved T cell responses and memory formation after OV therapies.^{32,44–48} In fact, many of the phenotypical changes observed in CAR T cells from each OV Responder group were also observed in Non-Responders, but to a lesser extent, except for CCR7 and TIGIT, which were significantly elevated in Goravir Responders compared to Non-Responders and CD56 and TIM-3 in R124 Responders compared to Non-Responders. This indicates that a successful response occurs through enhanced CAR T cell cytotoxicity and memory formation after interaction with OV-infected DMG cells.

Despite these positive observations, the absolute killing percentages observed *in vitro* used to define successful combinations, even though considered biologically meaningful, should not be directly equated with expected *in vivo* results or clinical efficacy. Both *in vivo* and therapeutic activity of the Ovs and the CAR T cells can be affected by additional factors not represented in our co-culture systems, including ineffective delivery, infiltration of OVs and CAR T cells into the tumor, neutralizing antibodies, CAR T cell expansion and persistence, cytokine-mediated effects, and TME remodeling. The latter could greatly affect the efficacy of CAR T cell therapy, as the brain TME contains a milieu of immunosuppressive cells.^{49,50} Our preclinical cultures fail to capture these effects; however, other preclinical models, such as patient-derived xenografts, also lack most of these populations and would thus fail to model innate or adaptive immune-mediated mechanisms.⁵¹ In addition, humanized mouse models cannot develop fully mature immune cells and are relatively costly for screening multiple patient-derived tumors.⁵² In this light, other *in vitro* investigations have used pe-

ripheral blood mononuclear cells from patients in co-culture screenings to partially alleviate this issue. In these co-cultures, treating tumor cells with OVs induced an increase in activation of T cells, NK cells, and a shift toward a pro-inflammatory phenotype of macrophages.^{53,54} However, these cultures did not incorporate the brain TME-exclusive immune population, microglia.⁵³ By using this *in vitro* setup, we aimed to assess CAR T cell-mediated DMG cytotoxicity in combination with a lytic virus, Goravir, and an immunogenic virus, R124, under defined OV concentrations, E:T ratios, and timing. These conditions allow us to determine the effects attributable solely to the interactions of CAR T cells with infected DMG cells, without other confounding factors that would otherwise be present in other models. Regardless, our work and previous investigations have demonstrated that infection of tumors with OVs can directly amplify CAR T cell cytotoxic capabilities.^{16,55} Although the presence of an intact TME would likely lead to higher combination efficacies, the observed therapeutic improvements and phenotypical changes in CAR T cells during combination with OVs in our study were nevertheless comparable to insights previously observed in *in vivo* studies.^{44–48} Future validation in a more immunologically representative model will still be required to further strengthen these findings.

In conclusion, our preclinical studies provide evidence for the ability of Goravir and R124 to expand the action of CAR T cells, improving anti-tumor killing by enhancing their effector phenotype. Despite the weaker oncolytic effect of R124 compared to Goravir, R124 infection of DMG cultures was more successful as a combination strategy. This highlights that immune stimulation is more important than oncolysis for the combination potential of OVs with immunotherapies. Finally, we demonstrate that the degree of OV-induced immune stimulation significantly affects anti-tumor killing, with Responders showing a stronger shift of CAR T cells toward a highly cytotoxic effector phenotype. These findings expand the possibility for new combinations of OVs and CAR T cells for the treatment of pediatric DMGs and identify OV-induced immune stimulation as a key driver of successful responses.

MATERIALS AND METHODS

Patient-derived cultures expansion and maintenance

VUMC-DIPG-10 (H3WT, DMG), VUMC-DIPG-G (H3.3K27M, DMG), and VUMC-DIPG-11 (H3.3K27M, DMG) were established from autopsy or resection material at Amsterdam UMC (Vrije University Medical Center of Amsterdam, Amsterdam, the Netherlands).⁵⁶ SU-DIPG-IV (H3.1K27M) was provided by Dr. Monje (Stanford University, Stanford, CA, USA)⁵⁷; HSJD-DIPG-07 (H3.3K27M, DMG) was received from Dr. Carcaboso (Hospital Sant Joan de Déu, Barcelona, Spain)⁵⁸; JHH-DIPG-01 (H3.3K27M, DMG) was provided by Dr. Raabe (John Hopkins Hospital, Baltimore, MD, USA).⁵⁹ All patient material was collected according to national and institutional guidelines and in accordance with the Declaration of Helsinki. Culture identities were confirmed through short tandem repeat analysis using the GenePrint 10 system (Promega, B9510, Leiden, the Netherlands),

and their negative mycoplasma status was confirmed using the MycoAlert Mycoplasma Detection kit (Lonza, LT07-318, Basel, Switzerland) prior to the initiation of the experiments. All cells were cultured and expanded as previously described.¹³

CAR T cells and UTD

B7H3- and GD2-CAR T cells were generated with a CD8 hinge—4-1BB—CD3z construct, including the scFV for B7H3 CAR (derived from patent: US20190153063) or the scFV for GD2 (derived from patent: WO2018118494), respectively, as previously described.⁶⁰ The CD4⁺ and CD8⁺ proportions, as well as CAR expression, were evaluated using fluorescence-activated cell sorting (FACS) after expansion, using CD3-APC (1:100, Biolegend, #300412), CD4 Violet Blue (1:50, Miltenyi biotec, #130-113-219), and CD8-V500 (1:50, BD, #561617), B7H3-CAR T cells were detected using a B7H3 detection reagent (ACRO Biosystems, B7B-HF2E7, DE, USA), and GD2-CAR T cells were detected using fluorescein isothiocyanate (FITC)-conjugated anti murine IgG (Sigma-Aldrich, #F9137, Amsterdam, the Netherlands). The analysis was performed on a Cytotrex S (Beckman Coulter, Woerden, the Netherlands) and analyzed using FlowJo (BD bioscience).

OVs

Goravir is a derivative of the human adenovirus-B AdV-lumc007, isolated from a gorilla modified with a 24-bp deletion of the Rb-binding domain of the E1A protein for selective replication in tumor cells.¹⁷ This mutant is capable of replication and was generated and titrated on HER911 cells as previously described.¹⁷ R124 is a wild-type strain of human reovirus type 3 Dearing (T3D) capable of replication in Rat sarcoma (RAS)-activated cells. R124 was generated and titrated on HER911 cells as previously described.¹⁸

CellTiter-Glo experiments

Single-cell suspensions of the cultures were prepared after treatment with Accutase (Merck, A6964-100ML, Amsterdam, the Netherlands), mechanical dissociation, and washing with phosphate buffered saline (PBS), followed by passage through a 70 µm strainer (Greiner bio-one, 542070, Alphen aan den Rijn, the Netherlands). Then, 2,000 cells per well of dissociated cultures or T cells were seeded onto 96-well U-bottom plates in 100 µL of complete tumor stem medium (TSM) and incubated at 37°C and 5% CO₂ for 24 h. Next, the seeded cells were infected with 100 µL of complete TSM containing increasing doses of the viruses and incubated at 37°C and 5% CO₂ for 72 h. At the end of the 72 h long incubation, CellTiter-Glo 3D reagent (Promega, G9681, Leiden, the Netherlands) was added to the wells and incubated for 10 min on an MTS 2/4 plate shaker (IKA, NC, USA) at room temperature (RT) and protected from light. Then, half of the volume from each well was transferred into new clear, flat-bottom black 96-well plates (Greiner, #655076, Mannheim, Germany), and luminescence was measured using the SpectraMax iD3 and the SoftMax Pro GxP software (Molecular devices, CA, USA). Viability was measured and normalized using the average values per culture or T cell values not infected with any viruses at the endpoint, which were set as 100%.

Immunofluorescent stainings

For imaging: DMG cultures were prepared as single-cell suspensions using Accutase and then resuspended in cold PBS +2% fetal calf serum (FCS) at a final concentration of 500.000 cells/mL. PBS was added to cytofunnels (Thermo Fisher Scientific, #11610591, NH, USA) attached and mounted on SuperFrost Plus glass slides (VWR international, #VWR 631-0108, PA, USA) with filter cards (Thermo Fisher Scientific, 11630601, NH, USA). The slides were centrifuged at 680 rcf for 3 min in order to properly pre-wet the filter cards. Then, 200 µL of cell suspension was added to the respective cytofunnels, spun at 680 rpm for 7 min, and left to air dry overnight at RT. The following day, slides were washed 3 times with PBS+0.1% Tween 20 (Sigma-Aldrich, #11332465001, MO, USA) (TPBS) and blocked using western blot blocking buffer (BB) for immunofluorescence (Rockland Immunochemicals, #MB-070, PA, USA) for 20 min at RT. Slides were then stained with anti-human B7H3(CD276)-APC (BioLegend, #351005, MIH42, CA, USA) and anti-human GD2-PE (BD Biosciences, #562100, 14.G2a, NJ, USA) in BB overnight at 4°C in the dark. Finally, slides were washed 3 times with Tris-buffered saline + 0.1% Tween 20 (TBST) and stained with Hoechst 3334 (Thermo Fisher Scientific, #62249, Amsterdam, the Netherlands) for 20 min at 4°C in the dark. Slides were washed 3 times with TBST +0.1% Tween 20 and air-dried in the dark at RT. Finally, images of the slides were acquired using a DMI Thunder Microscope (Leica Microsystems, Wetzlar, Germany) and the Leica Application Suite X (LAS X) program.

For FACS analysis, 20.000 cells/well of dissociated DMG cells prepared as described above, or CAR T and UTD cells, were seeded onto U-bottom 96-well plates in complete TSM. After 24 h of incubation at 37°C and 5% CO₂, complete TSM with 20/200 MOI Goravir, 10/100 MOI R124, or complete TSM alone was added to the appropriate wells and incubated for 24 or 72 h at 37°C and 5% CO₂. At each time point, the uninfected and infected DMG cultures, or CAR T and UTD cells, were collected and made into single-cell suspensions using Accutase for 5 min and then washed with PBS. For GD2 and B7H3 staining, cells were resuspended in a staining mix consisting of anti-human B7H3(CD276)-APC, anti-human GD2-PE, and LIVE/DEAD Fixable Aqua Dead Cell Stain Kit (Thermo Fisher Scientific, #L34957, Amsterdam, the Netherlands) in PBS for 20 min at 4°C in the dark. The cells were then washed and incubated with 1% paraformaldehyde (PFA) in PBS for at least 30 min at 4°C in the dark. For viral infectivity, the cells were washed with PBS and stained with the LIVE/DEAD Fixable Aqua Dead Cell Stain Kit in PBS for 20 min at 4°C. After staining, cells were washed with PBS and fixed in eBioscience fixation/permeabilization concentrate (Invitrogen, #00-5123-43, MA, USA) diluted 1:4 in eBioscience fixation/permeabilization diluent (Invitrogen, #00-5223-56) (fixation medium) for 15 min at 4°C. The fixed cells were subsequently washed and incubated in eBioscience permeabilization buffer (Invitrogen, #00-8333-56) for 20 min at 4°C. Then, the cells were washed with permeabilization buffer and stained for Goravir with anti-Hexon adenovirus type 3 (BioRad, #MCA489, CA, USA) in permeabilization buffer, and for R124 with anti-σ3 T3D reovirus antibody (DSHB, #4F2, IA, USA) in permeabilization buffer for 20 min at 4°C. After washing, cells

were stained with secondary antibodies conjugated with AF-568, anti-mouse IgG H&L (Abcam, #ab175700, Cambridge, UK) for 30 min at 4°C and washed once more in permeabilization buffer. Finally, the cells were washed and resuspended in PBS for measurement. FACS analysis was performed using a Cytotflex S. B7H3, GD2 expression, as well as anti-Hexon and anti-σ3, were measured after exclusion of dead cells (LIVE/DEAD negative selection) using Flowjo.

qPCR, bulk RNA isolation, and differential expression analysis

Total RNA from DMG cultures or CAR T and UTD cells infected with 20/200 MOI Goravir or 10/100 MOI R124 or treated with complete TSM for 0, 24, and 72 h was extracted using the RNeasy Mini kit (Qiagen, 74104, Venlo, the Netherlands) according to the manufacturer's protocol. For qPCR, the amount of extracted RNA was quantified using a NanoDrop Lite Plus (Thermo Fisher Scientific), followed by cDNA synthesis using the SuperScript III First-Strand Synthesis System (Invitrogen, #18080051). Finally, qPCR was performed using the PowerTrack SYBR Green Master Mix (Thermo Fisher Scientific, #A46109). All kits were used according to the manufacturer's instructions. The qPCR reaction was read using a CFX Connect Real-Time System (BioRad). The primers used in the experiment were as follows: (Goravir) adenovirus type B polymerase forward primer: 5'-ATGATGTCATAACCTGGTTGG-3' and reverse primer: 5'-TTACCGTGCAGACAAAGACG-3'; (R124) S4Q forward primer: 5'-CGCTTTTGAAGGTCGTGTATCA-3' and reverse primer: 5'-CTGGCTGTGCTGAGATTGTTT-3. For bulk RNA analysis, RNA quality and purity were assessed using the Agilent 2100 Bioanalyzer system with the Agilent RNA6000 Nano kit (Agilent, #5067-1511, CA, USA), and samples with an RIN value > 6.2 were included for poly(A) sequencing. All sequencing, adapter removal, trimming, mapping, and feature counting were performed by GenomeScan (Leiden, the Netherlands) using the human GRCh38.p13 genome as the reference. The raw count files were loaded in R and merged to create the expression matrix for all samples. Gene counts with the same gene symbol were aggregated using the R function "aggregate," and genes without an assigned gene symbol were removed. The resulting expression matrix was analyzed using the DESeq2⁶¹ R package. Genes with fewer than 10 counts and not expressed in at least six samples were filtered out. The filtered count matrix underwent variance stabilizing transformation for normalization, and the normalized matrix was used for PCA. For further visualization of cell identity-independent variation or heatmap generation, the cell identity batch effect was removed using "removeBatchEffect" from limma. Analysis of the data was performed using the limma package (v3.46.0; Ritchie et al., 2015). Differential expression analysis was performed using the raw filtered counts and a full design modeling the cell identity, the time point of collection, and the infection status of the samples using DESeq2. The uninfected samples were used as the reference for DEG comparisons using the default settings.

Reactome pathway analysis

Reactome pathway analysis was performed using the R package "ReactomPA"⁶² using the significant DEGs (adjusted *p* value < 0.05) and

ordered by log2 fold change. Reactome terms with an adjusted *p* value < 0.05 were considered significantly enriched by the DEGs in the corresponding groups.

Co-culture assays

As described above, dissociated and filtered DMGs were stained with 5 μM CellTraceViolet (CTV) (Thermo Fisher Scientific, #C34557) for 20 min at 37°C, 5% CO₂. After staining, the cells were washed with TSM medium for 5 min and then resuspended in complete TSM. Then, 20,000 cells/well of single-cell DMGs were seeded in a U-bottom 96-well plate in complete TSM. For the combination experiments 200 MOI of Goravir, 100 MOI of R124, or plain complete TSM were added to the indicated wells 24 h after seeding the DMG cells and incubated for an additional 24 h at 37°C and 5% CO₂. Thus, at 48 h after seeding the DMGs, CAR T cells or UTDs were added in complete TSM to the indicated wells at increasing E:T ratios and incubated for 48 h at 37°C and 5% CO₂. The plate was centrifuged at 300 rcf for 5 min at 4°C. For the measurement of tumor-specific killing, the cell pellet was collected, while the supernatants were collected for measuring cytokines and chemokines released.

For the measurement of anti-tumor killing, the collected cell pellets were resuspended and washed with cold PBS and then centrifuged. After discarding the supernatant, the cell pellets were dissociated with Accutase for 5 min at 37°C and 5% CO₂, followed by manual resuspension and washing with cold PBS. Next, the dissociated cell pellets were stained with 7-aminoactinomycin D (7-AAD) staining solution (Miltenyi Biotec, #130-111-568, Leiden, the Netherlands) in PBS for 20 min at 4°C in the dark. After washing, the cells were fixed in PBS with 1% PFA for at least 30 min at 4°C in the dark. Finally, the cells were washed and resuspended in PBS for measurement. Samples were analyzed by FACS using a Cytotflex S. The number of alive tumor cells (CTV+ and 7-AAD- cells) was calculated using FlowJo software. Cell killing was then normalized to the untreated control.

For the measurement of chemokine and cytokine release, the collected supernatants were left at RT to equilibrate in temperature. For measuring IFN-γ release, the Ready-Set-Go Human IFN-γ ELISA kit (# 88-7316-22, Thermo Fisher Scientific, Utrecht, the Netherlands) was used according to the manufacturer's instructions. The resulting absorbance was measured at 450 nm using the VersaMax plate reader (Molecular devices, CA, USA). Serial dilutions of IFN-γ standards were used to generate the standard curve and convert absorbance values into pg/mL. For measuring the remaining cytokines, supernatants from the co-cultures were transferred to V-bottom plates (BioLegend, #740379), and the LEGENDplex Mix and Match Human COVID-19 Cytokine Storm Panel 1&2 kit (BioLegend, #741095) was used following the manufacturer's protocol. More specifically, the chemokine/cytokine capture antibodies used for measurement were: Human IL-6 Capture Bead A3 (BioLegend, #741096), Human CCL2 (MCP-1) Capture Bead (BioLegend, #741097), Human IL-1β Capture Bead A5 (BioLegend, # 741115), Human IFN-α2 Capture Bead A6 (BioLegend, #741099), Human CCL5 (RANTES) Capture Bead A7

(BioLegend, #741109), Human sCD25 (IL-2Ra) Capture Bead A8 (BioLegend, #741117), Human IL-4 Capture Bead A10 (BioLegend, #741118), Human IL-7 Capture Bead B2 (BioLegend, #741102), Human IL-17 A Capture Bead B3 (BioLegend, #741120), Human CXCL8 (IL-8) Capture Bead B4 (BioLegend, #741104), Human TNF- α Capture Bead B5 (BioLegend, #741105), Human CXCL10 (IP-10) Capture Bead B6 (BioLegend, #741106), Human CCL3 (MIP-1 α) Capture Bead B7 (BioLegend, #741107), and Human IL-10 Capture Bead B9 (BioLegend, #741108). Fresh medium was used as a negative control during staining. The mean fluorescent intensity (MFI) of the different chemokines and cytokines was measured by FACS using a Cytoflex S and analyzed using the online BioLegend tool <https://legendplex.qognit.com/>. Finally, MFI values below the negative control MFI value were considered undetectable and were excluded from analysis.

CAR T cell phenotypical analysis using spectral cytometry

Single-cell suspension DMGs prepared as stated above were seeded in 96-well U-bottom plates at 50,000 cells per well and left to incubate in complete TSM for 24 h at 37°C and 5% CO₂. Then, 200 MOI of Goravir, 100 MOI of R124, or complete TSM were added to the indicated wells for an additional 24 h at 37°C and 5% CO₂. B7H3 and GD2 CAR T cells were then added at an E:T of 1:2 for B7H3 and 1:5 for GD2 in complete TSM in the indicated wells and incubated for 48 h at 37°C and 5% CO₂. At the end of the incubation period, the plates were centrifuged and the supernatants were discarded. The cell pellets were washed twice with PBS before live/dead staining with ViaDye Red (Cytek, R7-60008, Amsterdam, the Netherlands) in PBS for 20 min at 4°C in the dark. Then the cells were washed with PBS and then with FACS buffer (PBS +1% BSA +0.5 mM EDTA). The cells were then stained with a mix of antibodies (extracellular staining) (Table S5A) in FACS buffer and Brilliant stain buffer (BD Biosciences, 566349) for 20 min at 4°C in the dark. After staining, the cells were washed twice with FACS buffer and then fixed and permeabilized using the eBioscience Foxp3/Transcription Factor Staining Buffer Set (Invitrogen, 00-5523-00) (permeabilization buffer) for 30 min at 4°C in the dark. The fixed cells were washed twice with permeabilization buffer. The cells were then stained with a mix of antibodies (intracellular staining) (Table S5B) in permeabilization buffer for 20 min at 4°C in the dark. Finally, after washing with permeabilization buffer, the samples were further washed with FACS buffer. The samples were measured using a Cytek Aurora spectral cytometer. For spectral unmixing, single-stained beads (Cytek, #SKU B7-10011) for each antibody were used, and unstained samples were included as controls to remove autofluorescence per CAR T cell. The samples were unmixed using Spectroflo software (Cytek). Unmixed samples were analyzed in FlowJo, and population percentages and MFIs were measured in live (CD3+CD45+) T cells, excluding debris and DMG cells from the analysis.

Statistical analysis

All data are visualized as the mean $\pm$ standard deviation (SD). Multiple unpaired *t* test were used for parametric data, and

Mann-Whitney tests were used for non-parametric statistics of two-group comparisons. For multiple-group comparisons, paired one-way ANOVA corrected with Tukey's test was used for parametric data, while the Friedman paired test corrected with Dunn's test was used for non-parametric data. For comparisons of killing across monotherapies and combinations, two-way ANOVA corrected with Tukey's test was used. Differential expression and Reactome enrichment were performed in R using the default statistical and multiple correction models of the "DESeq2" and "ReactomPA" packages, respectively.

DATA AND CODE AVAILABILITY

All raw data and code used for the analysis, figure formation, and statistical analyses are freely available upon reasonable request. The bulk RNA sequencing data supporting the findings of this study are available in the GEO database under accession number GEO: GSE300244.

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AUTHOR CONTRIBUTIONS

Conceptualization and design, K.V., A.M.C., E.H., F.G.C., and J.v.d.L.; development of methodology, K.V., A.C.T.-C., P.W., A.M.C., M.v.H., N.M.M.D., C.F., V.K., R.C.H., S.N., J.v.d.L., E.H., and F.G.C.; acquisition of data and analysis, K.V., A.C.T.-C., L.P.J.v.O., and P.W.; writing—original draft, K.V., J.v.d.L., E.H., and F.G.C.; editing and reviewing of manuscript, K.V., A.C.T.-C., A.M.C., N.M.M.D., C.F., V.K., R.C.H., S.N., J.v.d.L., E.H., and F.G.C.; supervision, E.H., F.G.C., and J.v.d.L. All authors have read and agreed to the current version of the manuscript.

DECLARATION OF INTERESTS

R.C.H. is a named inventor on patents related to the use of Goravir as viral vectors or oncolytic agents. R.C.H. receives research funding from Janssen Vaccines & Prevention B.V. (Leiden, the Netherlands) for projects on adenoviruses. The funders had no role in the design of the study, the analyses or interpretation of data, the writing of the article, or the decision to publish the results.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.omton.2026.201191>.

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